



Predicting Excess Stock Returns with Earnings-Call Sentiment and Quantitative Factors

ML in Finance

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1 Introduction

Earnings conference calls are public information events in which managers discuss results, strategy, and outlook. Related financial-text evidence suggests that recent tone changes can carry more predictive content than sentiment levels alone, because levels are partly anticipated by the market [4]. We ask a precise question: **does fresh, firm-level earnings-call sentiment improve cross-sectional prediction of monthly excess stock returns beyond simple quantitative benchmarks, and if so, through which feature-engineering choices?**

We address this question using FinBERT [1] sentiment scores from 34,643 earnings calls matched to CRSP monthly returns over 2008–2023. Our contribution is primarily empirical and methodological: we document that signal freshness, quarter-over-quarter delta features, and cross-sectional standardisation together convert a weak raw sentiment signal into a modest, tradable cross-sectional predictor. We benchmark against 153 JKP factor-return signals [5] and compare XGBoost with a deep learning MLP across the same features.

Contributions

1. **A fresh FinBERT signal:** freshness filter (≤ 3 months), quarter-over-quarter delta features, and cross-sectional z-scoring. A preprocessing ablation shows each step is necessary; cross-sectional z-scoring is the most impactful for portfolio performance.
2. **Model comparison:** XGBoost and a PyTorch MLP on identical feature sets, plus sequence models (LSTM, Transformer). MLP results are consistent with a mildly non-linear signal; sequence models do not generalise across regimes.
3. **Economic evaluation:** long-short decile backtests with 0.2% one-way transaction costs, multi-period robustness, and a 108-month post-training IC analysis.

Main Findings

In our tests, the only signal that survives both the IC and backtest checks is fresh FinBERT sentiment. The FinBERT-only XGBoost long-short decile yields +11.4% net annualised return (Sharpe 0.79) over 2019–2023, with a Rank IC t -statistic of 3.79 over the 108-month post-training window. A cross-sectional MLP on the same features produces +7.87% net (Sharpe 0.58), so the signal is not tied to XGBoost alone. This is a modest signal, not a large forecasting edge.

JKP factor returns, being identical across all stocks within a given month, are structurally suited to time-series or market-timing applications rather than firm-level cross-sectional stock selection. Their poor long-short performance in our setup reflects a design mismatch, not a shortcoming of the JKP literature. The results are consistent with the hypothesis that fresh changes in earnings-call tone contain incremental information not fully captured by our benchmarks, though we caution that the test period (60 months) leaves meaningful statistical uncertainty.

2 Data

2.1 CRSP Monthly Returns

We use the CRSP monthly stock file for January 2008 to December 2023 (1,450,439 raw observations). The target variable is the monthly excess return:

$$r_{i,t}^e = \text{MthRet}_{i,t} - \text{sprtrn}_t$$

where sprtrn_t is the S&P500 return. Raw returns are winsorised at the 1st/99th percentiles. The excess return has mean -0.019% per month and 48.2% positive observations, confirming it is centred at zero.

2.2 JKP Quantitative Factor Returns

We use 153 long-short factor portfolio returns from [5] (momentum, value, quality, profitability, investment, low-risk). These are monthly time-series signals: each factor takes the same value for all stocks in a given month. **Design choice:** JKP factor returns for period t are realised over the same interval as CRSP returns; using them contemporaneously would be look-ahead. We lag them by one month before merging, so the vector at $t - 1$ is used to predict CRSP return at t .

This lagging implies that JKP features encode *market-level* information about recent factor performance. They may help with time-series allocation or regime identification, but they do not vary *across firms* within a given month and are therefore not naturally suited to cross-sectional stock selection. JKP factor returns are included here as a deliberately simple time-series benchmark, not as firm-level characteristics; any underperformance in our cross-sectional setup reflects a structural mismatch by design, not a shortcoming of the JKP factor library.

2.3 Earnings Calls and FinBERT Inference

We obtained 34,643 earnings call transcripts matched to CRSP firms by PERMNO. We extract the Q&A section (most forward-looking), chunk into 512-token segments, and score each chunk with ProsusAI/finbert [1], a BERT-based model fine-tuned on financial text. Each call yields three probabilities averaged over chunks: p_{pos} , p_{neg} , p_{neu} (summing to unity). Calls with fewer than 100 words are excluded. Inference details and score distributions are in Appendix A.

2.4 Feature Engineering

From the three raw FinBERT scores we construct the following features:

Table 1: Engineered FinBERT features. Delta features are NaN for a firm’s first earnings call.

Feature	Definition
net_sentiment	$p_{\text{pos}} - p_{\text{neg}}$: signed directional signal
sentiment_strength	$1 - p_{\text{neu}}$: non-neutrality of the call
delta_net	$\Delta \text{net_sentiment}$ quarter-over-quarter (<i>surprise signal</i>)
delta_pos, delta_neg	Quarter-over-quarter changes in individual components
months_since_call	Signal age (1, 2 or 3 by construction after freshness filter)
*_zscore	Cross-sectional z-score within each prediction month

Freshness filter. The raw merged dataset reveals a severe staleness problem: the median lag between the most recent earnings call and the prediction date is **7 months**, at which point the sentiment signal has largely decayed. We retain only stock-month pairs where the most recent call occurred within **3 months** of the prediction date, reducing the dataset from 517,776 to **122,847 observations**. Section 4.1 shows this filter is the most critical preprocessing step.

Universe note. The JKP-only model is evaluated on a broader universe (40,872 test observations) since it does not require earnings-call data. FinBERT-containing models are restricted to firms with a recent earnings call (≤ 3 months) and, for delta features, at least two consecutive calls (36,901 test observations). Results for FinBERT models therefore apply to an earnings-call-covered universe (predominantly mid- and large-cap firms). Cross-model comparisons within Table 3 are partially confounded by this universe difference.

Cross-sectional z-scoring removes time-varying means and scales, preventing the model from learning calendar effects that cannot be reproduced out-of-sample. **Delta features** are computed strictly using prior-quarter call data, with no look-ahead. The scaler is fitted on the training set only and applied to validation and test sets.

2.5 Temporal Split

We apply a strict temporal split with no random shuffling:

- **Train:** January 2008 – December 2014 (52,931 obs)
- **Validation** (early stopping only): January 2015 – December 2018 (29,044 obs)
- **Test** (all reported metrics): January 2019 – December 2023 (40,872 obs)

3 Methodology

3.1 Models

We train five model families targeting $r_{i,t}^e$, sharing the same temporal split and feature sets.

XGBoost variants (four models).

1. **JKP only**: 153 lagged factor features.
2. **FinBERT only**: 12 sentiment features (Table 1). *Primary model*.
3. **JKP + FinBERT**: all 165 features.
4. **JKP + FinBERT $\times 10$** : same 165 features, FinBERT columns weighted $10\times$ via `feature_weights` in `DMatrix`.

Shared hyperparameters: 500 estimators, early stopping at 30 rounds on validation RMSE, learning rate 0.05, max depth 4, subsample 0.8, L_2 regularisation 1.0, `random_state=42`.

Deep learning models (PyTorch). To verify that the signal generalises beyond a single model class, we train:

1. **MLP_{FinBERT}**: fully-connected MLP on the 12 FinBERT features. Architecture and training details in Appendix B.
2. **MLP_{JKP+FinBERT}**: MLP on all 165 features (neural-network equivalent of combined XGBoost).
3. **LSTM_{FinBERT}**: two-layer LSTM on the temporal sequence of the last 8 quarterly calls (Appendix B).
4. **Attention_{FinBERT}**: Pre-LN Transformer encoder on the same 8-step sequence (Appendix B).

All DL models use Adam ($\ell_r = 10^{-3}$), early stopping on validation Rank IC (patience 15, max 150 epochs), and Apple M4 MPS backend.

Simple linear benchmark. We additionally run a Ridge regression ($\alpha = 1.0$) on the full 12 FinBERT features to test whether the signal requires non-linear modelling.

3.2 Evaluation Metrics

- **Rank IC**: mean monthly Spearman correlation between predicted and realised excess returns, computed cross-sectionally within each month. This is the standard metric for cross-sectional equity prediction [2]. We distinguish this from global Rank IC (Spearman across all stock-month pairs), which can be inflated by time-series variation.
- **IC t -statistic**: $\bar{IC}/(\sigma_{IC}/\sqrt{T})$ over T monthly ICs.

3.3 Investment Strategies and Transaction Costs

Equal-weighted long-short decile strategy: long top 10%, short bottom 10%, rebalanced monthly. Transaction costs of **0.2% one-way** (0.4% round-trip) are applied on monthly turnover. We report annualised gross and net returns, Sharpe ratio, OLS alpha versus S&P500, and average monthly turnover.

4 Results

4.1 Feature Engineering Ablation

Table 2 isolates the contribution of each preprocessing step. Raw FinBERT levels alone are not exploitable (Sharpe -0.75). The freshness filter is the most critical step: retaining only calls within 3 months of the prediction date raises Sharpe to 0.28. Delta features incrementally improve Rank IC ($0.019 \rightarrow 0.026$), capturing the quarter-over-quarter surprise component. Cross-sectional z-scoring within each prediction month is the final and most impactful step (Sharpe $0.14 \rightarrow 0.79$): by ranking each firm relative to its peers in the same month, the model directly targets the cross-sectional objective of the long-short strategy. Ridge regression on the full feature set (Sharpe -0.10) performs poorly, suggesting the exploitable component of the signal is not well captured by a simple linear specification.

Table 2: Preprocessing ablation: FinBERT XGBoost, long-short decile, out-of-sample 2019–2023. Rank IC is the mean monthly cross-sectional Spearman. The full-setup row uses the saved FinBERT-only predictions (identical pipeline to Table 4).

Model	Feature set	Rank IC	Net (%)	Sharpe	Turn.
XGBoost	Raw levels ($p_{\text{pos}}, p_{\text{neg}}, p_{\text{neu}}$)	0.001	−7.30%	−0.75	49%
XGBoost	+ Freshness filter + aggregates	0.019	+3.47%	0.28	60%
XGBoost	+ Delta features (Δ net, Δ pos, Δ neg)	0.026	+2.08%	0.14	61%
XGBoost	+ Cross-sectional z-scores (full setup)	0.032	+11.40%	0.79	58%
Ridge	Full 12 features (linear benchmark)	0.002	−1.77%	−0.10	50%

4.2 Model Comparison (Unified Ablation)

Table 3 reports out-of-sample metrics on the 2019–2023 test set for all eight models.

Table 3: Unified ablation, out-of-sample 2019–2023 (long-short decile, 0.2% one-way costs). **Rank IC**: mean monthly cross-sectional Spearman, defined only where predictions vary across stocks. †: defined in 25% of months only; JKP features produce within-month constant predictions in the remaining 75%.

Family	Model	Rank IC	Net (%)	Sharpe	Turn.
XGBoost	JKP only	n/a*	−4.50	−0.36	91%
	FinBERT only	0.032	+11.4	0.79	58%
	JKP + FinBERT	0.023†	+0.49	0.04	91%
	JKP + FinBERT $\times 10$	−0.005	−8.53	−0.52	83%
Deep Learning	MLP _{FinBERT}	0.025	+7.87	0.58	67%
	MLP _{JKP+FinBERT}	−0.012	−12.9	−0.87	87%
	LSTM _{FinBERT}	−0.014	−2.31	−0.15	38%
	Attention _{FinBERT}	−0.013	+2.21	0.14	38%

* Within-month constant predictions: monthly cross-sectional Rank IC undefined.

Three observations hold across model families. First, the cross-sectional FinBERT signal is exploitable: FinBERT-only XGBoost (IC 0.032, Sharpe 0.79) and MLP_{FinBERT} (IC 0.025, Sharpe 0.58) both generate positive risk-adjusted returns, with XGBoost stronger on both metrics. Second, the JKP structural mismatch appears across all families: JKP-only predictions are within-month constant (monthly IC undefined), JKP+FinBERT is constant in 75% of months, and JKP+FinBERT $\times 10$ achieves IC −0.005 despite a high global IC inflated by cross-month variation. Third, Ridge on the same features yields IC 0.002 and Sharpe −0.10, suggesting the signal is not well captured by a linear specification, whereas both XGBoost and MLP extract positive signal.

4.3 Backtest Performance

Table 4 reports full backtest metrics for all models.

Table 4: Backtest results, long-short decile, out-of-sample 2019–2023, net of 0.2% one-way transaction costs.

Model	Gross	Net	Sharpe	Alpha	Beta	Turn.
JKP only	−0.12%	−4.50%	−0.36	−4.87%	0.03	91%
FinBERT only (XGBoost)	+14.2%	+11.4%	0.79	+10.6%	0.06	58%
JKP + FinBERT	+4.85%	+0.49%	0.04	+0.42%	0.01	91%
JKP + FinBERT $\times 10$	−4.54%	−8.53%	−0.52	−8.07%	−0.03	83%
MLP _{FinBERT}	+11.1%	+7.87%	0.58	+6.24%	0.11	67%
MLP _{JKP+FinBERT}	−8.70%	−12.9%	−0.87	−14.7%	0.13	87%
LSTM _{FinBERT}	−0.49%	−2.31%	−0.15	−2.43%	0.01	38%
Attention _{FinBERT}	+4.02%	+2.21%	0.14	+1.82%	0.03	38%

FinBERT-only XGBoost (Sharpe 0.79, Alpha +10.6%) and $\text{MLP}_{\text{FinBERT}}$ (Sharpe 0.58, Alpha +6.2%) are the only two models with both positive monthly IC and positive net returns. XGBoost achieves both higher IC (0.032 vs 0.025) and higher Sharpe, making it the stronger model in our setup; the MLP result shows the signal is not tied to XGBoost alone. Sequence models (LSTM, Attention) are discussed in Section 5. The Attention model’s positive net return (+2.21%) despite negative monthly IC (−0.013) is explained by compounding: very strong early performance (2019–2020 Sharpe >1.0) offsets sustained 2021–2023 losses in dollar-weighted terms even as the equal-weight average IC turns negative.

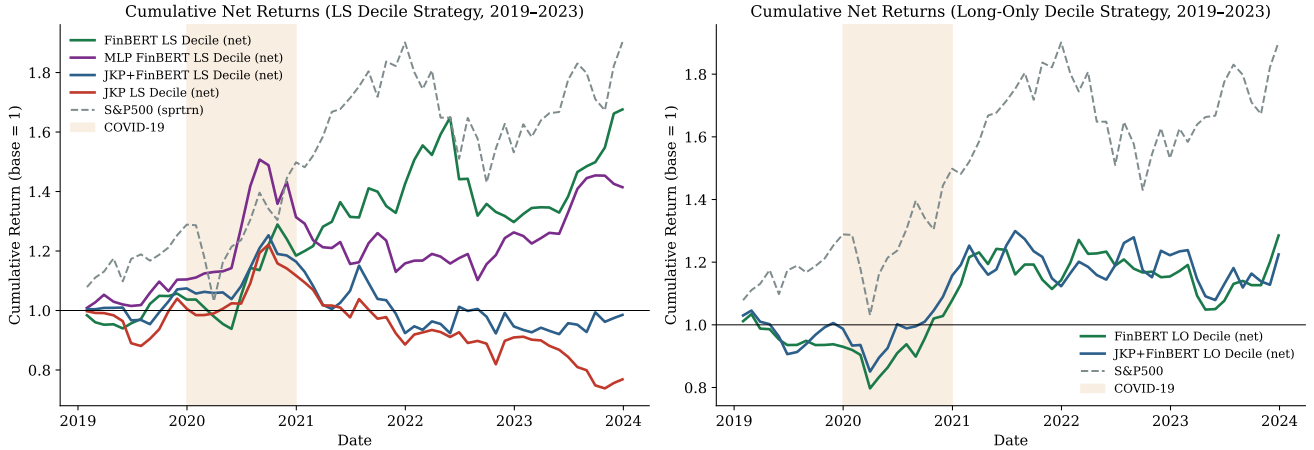


Figure 1: Cumulative net returns: LS Decile (left) and LO Decile (right), 2019–2023. S&P500 dashed benchmark. Orange band: COVID-19 period.

4.4 Signal Persistence and Robustness

Post-training IC path (108 months). Using the model trained on 2008–2014 with early stopping monitored on 2015–2018, we evaluate the full 2015–2023 post-training prediction path. FinBERT-only achieves mean monthly IC 0.0290, positive in 68% of months, and IC t -statistic 3.79. Because 2015–2018 is also the early-stopping window, the formal untouched test period remains 2019–2023; the 108-month path is used as a persistence diagnostic. JKP-only yields undefined monthly IC (within-month constant predictions). Full rolling results are in Appendix E.

Sub-period performance. FinBERT-only is profitable in all three sub-periods: pre-COVID 2019 (+3.87%, Sharpe 0.50), COVID 2020 (+14.99%, Sharpe 0.77), and post-COVID 2021–2023 (+12.65%, Sharpe 0.87). The strongest sub-period is *after* the COVID-19 shock, suggesting performance is not driven by a single episode. Transaction cost sensitivity (0.1%–0.4% one-way) is reported in Appendix D.

5 Discussion

5.1 JKP Factor Returns: A Structural Mismatch, Not a Failure

JKP factor returns are market-level, time-series signals: they take the same value for all stocks within a given month, encoding information about recent factor-portfolio performance. This structure is well suited to time-series allocation or regime identification, but not to firm-level cross-sectional stock selection.

In a long-short portfolio, returns are measured as long minus short *within* the same month. A prediction that is constant across stocks in a given month produces a random within-month ranking, yielding zero expected return before costs and negative net returns after. This explains why JKP-only yields undefined monthly Rank IC and −4.5% net annualised return: not because JKP factors lack information, but because that information is aligned with time-series variation rather than firm-level cross-sectional differences.

The *global* Rank IC (Spearman across all stock-month pairs) can appear positive for JKP models because it captures cross-month variation: correctly calling “good” versus “bad” months inflates global IC even when within-month stock rankings are uninformative. The combination paradox (JKP + FinBERT has higher global IC than FinBERT alone yet worse portfolio performance) follows directly: added JKP features capture time-series variation that inflates global IC but degrades cross-sectional discrimination, while also raising portfolio turnover from 58% to 91% ($\approx 3\%$ additional annual transaction costs). The same mismatch appears across both XGBoost and deep learning models.

5.2 XGBoost vs. Deep Learning

XGBoost vs. MLP. On the same 12 FinBERT features, XGBoost achieves monthly IC 0.032 and Sharpe 0.79, versus IC 0.025 and Sharpe 0.58 for $\text{MLP}_{\text{FinBERT}}$. Both models generate positive cross-sectional IC and profitable backtests, suggesting that the signal is not tied to XGBoost alone. XGBoost performs best in this setup, likely because it handles the irregular sparsity of sentiment features more efficiently through tree-based splits. The Ridge linear benchmark (IC 0.002, Sharpe -0.10) performs poorly, suggesting the exploitable component of the signal is not well captured by a simple linear specification.

Sequence models do not generalise across regimes. $\text{LSTM}_{\text{FinBERT}}$ and $\text{Attention}_{\text{FinBERT}}$ model the temporal trajectory of a firm’s earnings calls over up to 8 quarters. Both models perform well in 2019–2020 but deteriorate sharply in 2021–2023 (see Appendix E). The quarterly trajectory patterns learned on 2008–2018 data do not transfer to the post-COVID rate-hike regime. Sequence models are less robust because quarterly earnings-call trajectories are sparse and regime-dependent; the cross-sectional rank of current sentiment is more regime-stable.

Economic interpretation. The results suggest that what matters is *how a firm’s tone ranks against peers today*, not *how its tone has evolved over the past two years*. This is consistent with findings in the related analyst-report literature [4], where recent tone changes carry more predictive content than multi-period sentiment histories.

5.3 Limitations

1. **Universe:** FinBERT models cover only earnings-call firms ($\approx$ mid- and large-cap). Results may not generalise to small-cap or infrequently covered stocks.
2. **Statistical power:** five years (60 months) of untouched test data leave meaningful uncertainty around the Sharpe estimate; the 108-month post-training IC path ($t = 3.79$) is a useful persistence check, but 2015–2018 is also used for early stopping.
3. **Freshness cutoff:** the 3-month threshold was set a priori. Sensitivity analysis (Appendix C) shows IC at 1-month cutoff is $3\times$ higher; the optimal cutoff for risk-adjusted backtest performance remains untested.
4. **Transaction costs:** 0.2% one-way is conservative for large-caps but may underestimate costs for smaller names. Sensitivity analysis (Appendix D) shows FinBERT-only remains profitable up to 0.4% one-way.
5. **JKP comparison:** JKP-only is evaluated on a broader universe (40,872 test obs.) since it does not require earnings calls; FinBERT models require a recent call (≤ 3 months) and, for delta features, at least two consecutive calls (36,901 test obs.). This partially confounds cross-model comparisons.
6. **FinBERT context window:** the 512-token limit requires chunking; very long Q&A sections may lose information in later chunks.

6 Conclusion

Fresh, firm-level earnings-call sentiment does improve cross-sectional prediction of monthly excess stock returns, though modestly. The preprocessing ablation (Table 2) shows that freshness filtering, delta features, and cross-sectional z-scoring are each necessary: freshness filtering produces the first profitable baseline and z-scoring lifts Sharpe from 0.14 to 0.79. Over the 2019–2023 test period, FinBERT-only XGBoost achieves +11.4% net annualised return (Sharpe 0.79). As a persistence diagnostic, the IC t -statistic over the full 108-month post-training path (2015–2023) is 3.79; because 2015–2018 also serves as the early-stopping window, the formal untouched test period remains 2019–2023. A cross-sectional MLP on the same features produces +7.87% net (Sharpe 0.58), and a Ridge baseline (Sharpe -0.10) suggests the relationship is not well captured by a simple linear specification.

JKP factor returns are not well aligned with this cross-sectional strategy design: being constant across stocks within a month, they encode market-level time-series information rather than firm-level cross-sectional differences. Sequence models (LSTM, Transformer) are not robust out-of-sample because quarterly earnings-call trajectories are sparse and regime-dependent.

The signal is modest (monthly IC ≈ 0.03), the test period limited (60 months), and the universe skewed toward mid- and large-cap firms with regular earnings calls. Still, the evidence points toward fresh tone changes carrying incremental information not captured by our quantitative benchmarks. Higher-frequency updates, alternative language models, or broader universe coverage are natural directions for follow-up.

References

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Appendix

A. FinBERT Inference Details

FinBERT inference ran on an Apple M4 GPU (MPS backend, ≈ 3 hours for 34,643 calls). Each transcript’s Q&A section is extracted via regex and chunked into 512-token segments. Scores (p_{pos} , p_{neg} , p_{neu}) are averaged over chunks weighted equally. Calls with fewer than 100 words are excluded. Figure 2 shows the score distributions: neutral sentiment dominates (mean $p_{\text{neu}} = 0.71$), with $p_{\text{pos}} = 0.24 \gg p_{\text{neg}} = 0.05$, consistent with the well-documented positivity bias in managerial communications.

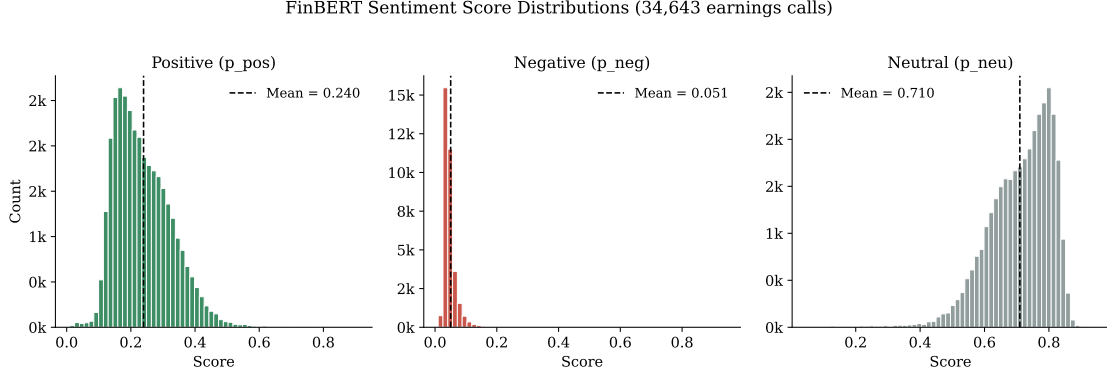


Figure 2: FinBERT sentiment score distributions (34,643 earnings calls).

B. Deep Learning Architecture Details

All DL models use Adam ($\ell_r = 10^{-3}$, weight decay 10^{-4} , batch size 512, gradient clipping 1.0), early stopping on validation Rank IC (patience 15, max 150 epochs), and `ReduceLROnPlateau` (factor 0.5, patience 5).

1. **MLP_{FinBERT}**: $12 \rightarrow 64 \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Drop}(0.3) \rightarrow 32 \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Drop}(0.3) \rightarrow 1$.
2. **MLP_{JKP+FinBERT}**: $165 \rightarrow 128 \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Drop}(0.3) \rightarrow 64 \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Drop}(0.3) \rightarrow 1$.
3. **LSTM_{FinBERT}**: $\text{LSTM}(3 \rightarrow 32, 2 \text{ layers}) \rightarrow \text{Drop}(0.3) \rightarrow \text{Linear}(32 \rightarrow 1)$. Input: last 8 quarterly call scores ($p_{\text{pos}}, p_{\text{neg}}, p_{\text{neu}}$), left-padded with zeros.
4. **Attention_{FinBERT}**: $\text{proj}(3 \rightarrow 32) + \text{PosEnc} \rightarrow 2 \times \text{EncoderLayer}(d=32, h=4, \text{ff}=64, \text{Pre-LN}) \rightarrow \text{MeanPool} \rightarrow \text{Linear}(32 \rightarrow 1)$. Padding mask applied; mean-pooled over non-padded positions.

C. Freshness Cutoff Sensitivity

Table 5: Freshness cutoff sensitivity: JKP + FinBERT XGBoost, global Rank IC (Spearman across all stock-month pairs). Values are shown for relative comparison only and are not comparable to the monthly cross-sectional IC in the main tables.

Cutoff	N train	N test	Rank IC
≤ 1 month	17,659	16,158	0.144
≤ 2 months	29,153	26,378	0.028
≤ 3 months	40,696	36,901	0.051

The 1-month cutoff yields IC of 0.144, roughly $3\times$ the 3-month value. The signal decays rapidly: the information advantage dissipates within 2–3 months, consistent with gradual price discovery [3].

D. Transaction Cost Sensitivity

Table 6: Transaction cost sensitivity: long-short decile, 2019–2023.

Model	0.1%		0.2% (base)		0.3%		0.4%	
	Net	SR	Net	SR	Net	SR	Net	SR
FinBERT (XGBoost)	+12.8%	0.88	+11.4%	0.79	+10.0%	0.69	+8.61%	0.60
MLP _{FinBERT}	+9.48%	0.69	+7.87%	0.58	+6.26%	0.46	+4.65%	0.34

Both models remain profitable across the full cost range. FinBERT-only retains Sharpe > 0.6 even at 0.4% one-way.

E. Rolling IC and Sub-Period Analysis

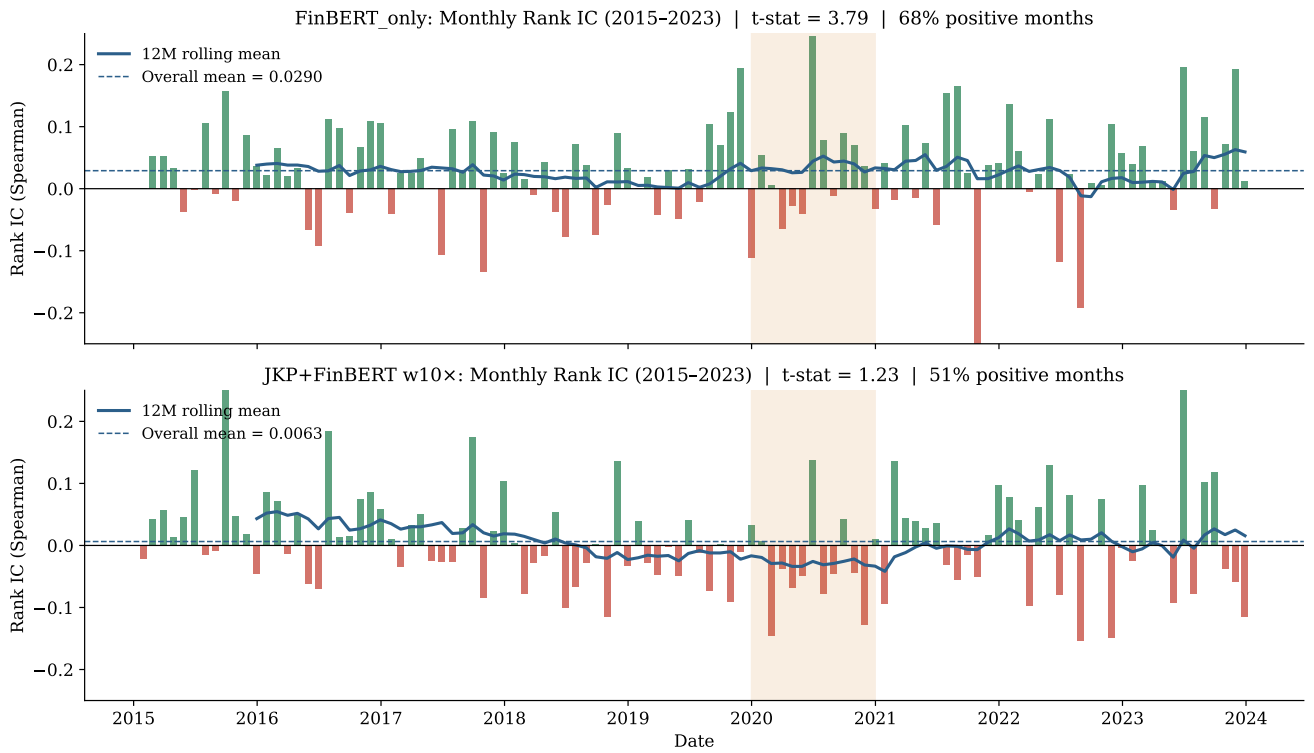


Figure 3: Monthly Rank IC for FinBERT-only (top) and JKP+FinBERT $\times 10$ (bottom), 2015–2023.

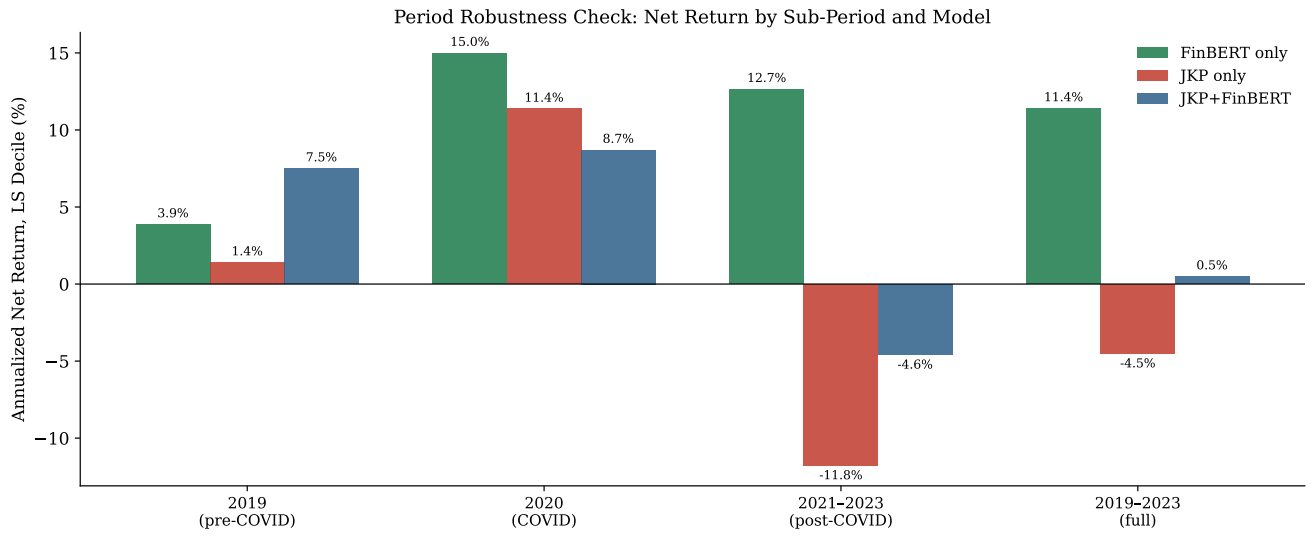


Figure 4: Net return by sub-period. FinBERT-only is positive in all three sub-periods; its best period is 2021–2023, after the COVID-19 shock.

Table 7: FinBERT-only XGBoost LS Decile: net return and Sharpe by sub-period.

Period	Net	Sharpe	Remark
2019 (pre-COVID)	+3.87%	0.50	Positive in normal conditions
2020 (COVID-19)	+14.99%	0.77	Strong but not unique driver
2021–2023 (post-COVID)	+ 12.65%	0.87	Best sub-period
2019–2023 (full)	+11.40%	0.79	Full out-of-sample window